Katie Matton

kmatton@mit.edu · kmatton.github.io · Google Scholar

Interests

machine learning, causality, language modeling, explainability, health applications

Education

Massachusetts Institute of Technology 2020 – April 2027 (expected)
Ph.D. in Electrical Engineering and Computer Science
Advisors: John Guttag and Rosalind W. Picard

University of Michigan, Ann Arbor 2018 – 2019
M.S.E. in Computer Science and Engineering
Advisor: Emily Mower Provost

University of Michigan, Ann Arbor 2014 – 2018
B.S.E. in Computer Science and Engineering, *summa cum laude*
Minor in Mathematics

Honors & Awards

Spotlight Paper (top 3% of submissions), ICLR	2025
Top Reviewer Award, Machine Learning for Health (ML4H)	2022, 2025
MIT Abdul Latif Jameel Clinic (J-Clinic) Fellowship	2020 – 2021
NSF Graduate Research Fellowship, Honorable Mention	2020
James B. Angell Scholar, University of Michigan	2018, 2019
Innovation Award, Michigan Engineering Graduate Research Symposium	2018
Best Research Poster, Michigan AI Symposium	2018
Engineering Honors Program, University of Michigan	2016 – 2018
D.E. Shaw Discovery Fellowship	2016

Publications

Preprints

* denotes equal contribution

Counterfactual Test-Time Augmentation for Shortcut Removal in Language Models
Katie Matton, Rosalind W. Picard, John Guttag, Maggie Makar
arXiv:2610.33904, 2026. Under review.

MedFuzz: Exploring the Robustness of Large Language Models in Medical Question Answering
Robert Ness, **Katie Matton**, Hayden Helm, Sheng Zhang, Junaid Bajwa, Carey E. Priebe, Eric Horvitz
arXiv:2406.06573, 2024.

Conference and Journal Papers

Classifying Phonotrauma Severity from Vocal Fold Images with Soft Ordinal Regression
Katie Matton, Purvaja Balaji, Hamzeh Ghasemzadeh, Jameson C. Cooper, Daryush D. Mehta, Jarrad H. Van Stan, Robert E. Hillman, Rosalind W. Picard, John Guttag, S. Mazdak Abulnaga
Machine Learning for Health (ML4H), 2025.

Walk the Talk? Measuring the Faithfulness of Large Language Model Explanations
Katie Matton, Robert Ness, John Guttag, Emre Kıcıman
International Conference on Learning Representations (ICLR), 2025. ***Spotlight*** *(top 3%).*

Contrastive Learning of Electrodermal Activity Representations for Stress Detection
Katie Matton[*], Robert Lewis[*], John Guttag, Rosalind W. Picard
Conference on Health, Inference, and Learning (CHIL), 2023.

Understanding the Impact of COVID-19 on Online Mental Health Forums
Laura Biester, **Katie Matton**, Janarthanan Rajendran, Emily Mower Provost, Rada Mihalcea
ACM Transactions on Management Information Systems, 2021.

When to Intervene: Predicting Individual Outcomes for Rapid Intervention (PRIORI) Using Passively Collected Speech from Smartphone Conversations
John Gideon, **Katie Matton**, Steve Anderau, Melvin G. McInnis, Emily Mower Provost
Bipolar Disorders, 2020.

Into the Wild: Transitioning from Recognizing Mood in Clinical Interactions to Personal Conversations for Individuals with Bipolar Disorder
Katie Matton, Melvin G. McInnis, Emily Mower Provost
Interspeech, 2019. ***Oral presentation****.*

Workshop Papers

A Removal Based Approach to Improve LLM Faithfulness at Test-Time
Qinglan Luo, S M A Nahian, John Guttag, S. Mazdak Abulnaga, **Katie Matton**
Workshop on Trustworthy AI for Good, NeurIPS 2026.

Improving Domain Generalization in Contrastive Learning using Adaptive Temperature Control
Katie Matton[*], Robert Lewis[*], Rosalind W. Picard, John Guttag
Workshop on Self-Supervised Learning, NeurIPS 2023.

Invariance-Based Causal Estimation in the Presence of Concept Drift
Katie Matton, John Guttag, Rosalind W. Picard
Workshop on Causal Representation Learning, UAI 2022.

Quantifying the Effects of COVID-19 on Mental Health Support Forums
Laura Biester[*], **Katie Matton**[*], Janarthanan Rajendran, Emily Mower Provost, Rada Mihalcea
Workshop on NLP for COVID-19, EMNLP 2020. ***Oral presentation****.*

Invited Talks

Walk the Talk? Measuring the Faithfulness of Large Language Model Explanations
Interpretability Special Interest Group, Google DeepMind (virtual) 2025

Contrastive Learning of Electrodermal Activity Representations for Stress Detection
Department of Biomedical Informatics, Emory University and Georgia Tech (virtual) 2023

Machine Learning for Mental Health Monitoring
Artificial Intelligence Journal Club, AstraZeneca (virtual) 2023

My Research Journey at the Intersection of Machine Learning and Behavioral Health
Beginners in Machine Learning Research group, Cohere for AI (virtual) 2023

Contrastive Learning for Wearable Sensor Data
Multidisciplinary Institute in AI (MIAI), Université Grenoble Alpes, Grenoble, France 2022
Digital Mental Health Workshop, MIT Media Lab 2022

Machine Learning for Mental Health
Media Lab Focus Series, MIT 2021

Contributed Talks & Posters

Walk the Talk? Measuring the Faithfulness of Large Language Model Explanations
International Conference on Learning Representations (ICLR), Singapore. Poster. 2025
Workshop on Reliable and Responsible Foundation Models, ICLR 2024. Poster. 2024
Workshop on Secure and Trustworthy Large Language Models, ICLR 2024. Poster. 2024

Improving Domain Generalization in Contrastive Learning using Adaptive Temperature Control
Workshop on Self-Supervised Learning, NeurIPS 2023, New Orleans, LA. Poster. 2023
Workshop on Distribution Shifts, NeurIPS 2023, New Orleans, LA. Poster. 2023

Contrastive Learning of Electrodermal Activity Representations for Stress Detection
Conference on Health, Inference, and Learning (CHIL), Cambridge, MA. Poster. 2023
Workshop on Learning from Time Series for Health, NeurIPS 2022. Poster. 2022

Invariance-Based Causal Estimation in the Presence of Concept Drift
Workshop on Causal Representation Learning, UAI 2022, Eindhoven, Netherlands. Poster. 2022

Quantifying the Effects of COVID-19 on Mental Health Support Forums
Workshop on NLP for COVID-19, EMNLP 2020 (virtual). Oral. 2020

Into the Wild: Transitioning from Recognizing Mood in Clinical Interactions to Personal Conversations for Individuals with Bipolar Disorder
Interspeech 2019, Graz, Austria. Oral. 2019
Michigan AI Symposium. Poster. 2019

Mood State Prediction for Individuals with Bipolar Disorder
Young Female Researchers in Speech Workshop, Interspeech 2018, Hyderabad, India. Poster. 2018
Michigan Engineering Graduate Research Symposium (Innovation Award). Poster. 2018
Michigan AI Symposium (Best Research Poster). Poster. 2018

Experience

Microsoft Research, Redmond, WA Summer 2023
Research Intern, Causal Machine Learning Team · Mentors: Emre Kıcıman and Robert Ness
Developed a method for measuring the faithfulness of LLM explanations and contributed to MedFuzz, a method for testing the robustness of medical LLMs.

University of New South Wales, Sydney, Australia January 2020
Visiting Research Assistant, Speech Processing Lab · Advisor: Vidhyasaharan Sethu
Examined methods for quantifying uncertainty in neural networks applied to depression severity prediction.

University of Michigan, Ann Arbor, MI 2017 – 2020
Research Assistant, Computational Human AI Lab · Advisor: Emily Mower Provost
Developed methods for speech- and language-based detection of mood states in individuals with bipolar disorder.

Microsoft, Redmond, WA Summers 2016, 2017
Software Engineering Intern
Built algorithms for measuring the quality of responses to location-based queries produced by Cortana, Microsoft's virtual assistant.

Teaching

Massachusetts Institute of Technology

Co-Instructor, 6.S091 Causality, with Chandler Squires and Jiaqi Zhang — January 2024
Teaching Assistant, 6.S091 Causality — January 2023
Teaching Assistant, 6.C51 Modeling with Machine Learning — Spring 2023

University of Michigan

Teaching Assistant, EECS 281 Data Structures and Algorithms — 2017 – 2018
Proof Tutor, Math 217 Linear Algebra — 2016 – 2017

Mentoring

Purvaja Balaji 2024 – 2025
M.Eng. thesis, MIT · Co-supervised with John Guttag and S. Mazdak Abulnaga
Phonotrauma severity classification from vocal fold images; published at ML4H 2025.

Qinglan Luo and S M A Nahian 2025 – present
Undergraduate researchers (UROP), MIT · Co-supervised with S. Mazdak Abulnaga
Test-time methods for improving LLM explanation faithfulness; accepted at the NeurIPS 2026 Workshop on Trustworthy AI for Good.

Professional Service

Reviewing

Neural Information Processing Systems (NeurIPS) — 2026
International Conference on Learning Representations (ICLR) — 2025
Machine Learning for Health (ML4H) — 2022, 2024, 2025
Workshop on Self-Supervised Learning, NeurIPS — 2023

Leadership and Organizing

Junior Chair, Causality Roundtable, ML4H Research Roundtables — 2023, 2025
Organizer, Causal Representation Learning Reading Group, MIT — 2022 – 2023
Co-President, Graduate Women in Course 6 (EECS), MIT — 2021 – 2022
Team Lead, Better Living Using Engineering (BLUE) Lab — 2014 – 2017
Led a 30+ member University of Michigan student team on co-designing engineering projects for health and education with partner communities.

Mentoring and Outreach

Panelist, Graduate Student Panel for Undergraduate Researchers, MIT — 2025
Mentor, Graduate Application Assistance Program, MIT EECS — 2020 – 2024
Mentor, AI Mentorship Program, MIT — 2020 – 2021
Instructor, Girls Encoded middle school outreach, Ann Arbor, MI — 2018 – 2019
Speaker and Instructor, AI Family Festival, Detroit, MI — 2018
Organizing Committee, Elementary Engineering Education Day, Ann Arbor, MI — 2015

Other

Boston Marathon finisher, 2:58:32 — 2026
Boston Marathon finisher, 3:06:23 — 2025